Example 5.16

The following data are the weights of students in a class. Find the median weights of the students

Weight(kg)	10	20	30	40	50	60	70
Number of Students	4	7	12	15	13	5	4

Solution:

Weight (kg) x	Frequency f	Cumulative Frequency $c.f$
10	4	4
20	7	11
30	12	23
40	15	38
50	13	51
60	5	56
70	4	60
Total	N = 60	

Here, $N = \sum f = 60$

$\frac{N+1}{2} = 30.5$

The cumulative frequency greater than 30.5 is 38.The value of x corresponding to 38 is 40. The median weight of the students is 40 kgs

(c) Median for Continuous grouped data

In this case, the data is given in the form of a frequency table with class-interval etc., The following formula is used to calculate the median.

$$\text{Median} = l + \frac{\frac{N}{2} - m}{f} \times c$$

Where

l = Lower limit of the median class

N = Total Numbers of frequencies

f = Frequency of the median class

NOTE

If the class intervals are given in inclusive type, convert them into exclusive type and call it as true class interval and consider the lower limit in it.

Ch5_11th Stats.indd 128 2/3/2019 11:00:58 AM